
Multimodal Signal Fusion for Cross-Sectional Equity Returns: Prediction, Implementation, and Capacity

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Abstract

We study whether combining chart-based and structured financial signals improves weekly cross-sectional equity portfolios, whether state-sensitive fusion adds value beyond simpler ensembles, and which gains survive trading frictions and scale. Using a common 2001–2024 stock-date universe and leakage-safe walk-forward forecasts, we compare a chart-image CNN, a structured-context random forest (RF), a 50/50 rank blend, a walk-forward logistic stack, a hidden Markov model (HMM), and an expected-return fusion rule. Fusion is valuable, but the HMM does not outperform simpler alternatives on a gross basis: the untuned rank blend has the highest gross Sharpe. The HMM instead combines lower portfolio risk than RF with substantially more stable rankings than the simpler fusion rules. In portfolios that account for existing holdings, transaction costs, and trading capacity, HMM has the highest net Sharpe at low AUM, while RF becomes superior as capital increases. HMM’s higher turnover becomes increasingly costly with scale. The same short sleeve that accounts for most of HMM’s lower portfolio volatility is also its main implementation bottleneck because turnover there is substantially more expensive. Finally, changes in the HMM’s inferred state are strongly associated with portfolio rebalancing and turnover, but not with incremental returns. The evidence supports multimodal fusion while showing that implementation stability and capacity can matter as much as gross predictive strength.

1 Introduction

Technical analysis remains widely used despite longstanding skepticism about whether price histories contain reliable predictive information. Earlier work finds limited short-run predictability in technical patterns (Lo, Mamaysky, and Wang 2000), while more recent research shows that convolutional neural networks (CNNs) trained on stock-chart images can learn predictive signals directly from prices and volume (Jiang, Kelly, and Xiu 2023).

A chart-based CNN, however, remains a chart-only model. Investors also observe structured information such as accounting variables, recent returns, liquidity, and macroeconomic conditions. Chart images and structured financial variables may therefore contain complementary information, creating a natural model-combination problem: how should signals from the two sources be combined, and does greater flexibility in the fusion rule remain valuable once trading costs and capacity are considered?

We study these questions using a chart-based CNN and a random forest (RF) trained on structured financial information. We compare four methods for combining their forecasts: an untuned 50/50 rank blend, a walk-forward logistic stack, a state-sensitive hidden Markov model (HMM), and an expected-return fusion rule. All models are evaluated on the same 2001–2024 stock-date universe using leakage-safe walk-forward forecasts and point-in-time data controls. The economic evaluation forms weekly long–short portfolios from the highest- and lowest-ranked stocks and then accounts for existing holdings, transaction costs, nonlinear market impact, and ADV-based trading constraints.

The empirical results indicate that the two modalities contain complementary return information. Spanning regressions show that HMM earns statistically significant alpha relative to RF alone and relative to RF and CNN jointly. The evidence for a distinct HMM return component weakens once the simpler fusion benchmarks are introduced: its alpha is no longer statistically distinguishable from zero relative to the 50/50 rank blend, logistic

stack, or expected-return fusion. Greater model complexity therefore does not monotonically improve gross performance, and the untuned 50/50 rank blend delivers the highest gross Sharpe. HMM instead occupies an intermediate position, combining lower portfolio risk than RF with greater ranking stability than the other fusion methods. The resulting risk–stability trade-off becomes economically important after costs. HMM has the highest net Sharpe at low AUM, whereas RF becomes superior from roughly \$50 million onward because its more persistent rankings require less trading.

RF overtakes HMM as AUM rises primarily because of implementation costs. HMM trades more than RF, and those additional trades become increasingly expensive as AUM increases; nonlinear market impact accounts for most of the resulting cost gap. The short sleeve is central to this trade-off: it accounts for most of HMM’s lower portfolio variance, but short-side trading is roughly twice as costly relative to turnover as long-side trading. Changes in the dominant HMM state are also associated with larger shifts in portfolio composition and higher turnover, but not with higher incremental returns. Taken together, these results suggest that the latent states primarily change how the two signals are combined and expressed in the portfolio rather than identifying periods in which the HMM earns superior returns.

Related work is summarized in Appendix A. Our contribution is primarily empirical: we show that chart-based and structured signals contain complementary return information and that combining them can deliver statistically significant incremental performance relative to the standalone experts. We then separate those gross forecasting gains from implementability, identify how the ranking of fusion methods changes with AUM, and trace that reversal to signal persistence, trading costs, and sleeve-level risk.

2 Data and Experimental Design

The visual branch transforms daily OHLCV histories into stock-chart images following the normalization logic of Jiang, Kelly, and Xiu (2023), while the structured branch uses a random forest (RF) trained on point-in-time accounting, return-based, and macroeconomic information drawn primarily from CRSP, WRDS/Compustat, and Federal Reserve data, following the broader real-time structured-information design of Bastianello (2026). The CNN predicts the direction of the next five-trading-day return, while the RF predicts the direction of the next 20-trading-day return and serves as a medium-horizon structured-context expert. The final fusion target and portfolio evaluation are based on the five-day horizon. We evaluate the CNN and RF as standalone signals together with four fusion approaches: a 50/50 rank blend, a walk-forward logistic stack, a hidden Markov model (HMM), and an expected-return fusion rule. Detailed data construction and preprocessing are reported in Appendix C.1; the models themselves are described in Section 3.

All headline comparisons use a common stock-date overlap universe. Base-model history begins in 1993, and the reported out-of-sample portfolio evaluation covers 2001–2024. Forecasts and annual model-selection decisions use only information available before the prediction date. The tradable evaluation applies price and dollar-volume screens; screened-versus-unscreened results are retained as a robustness exercise rather than as the central result.

We evaluate each signal by its cross-sectional ranking. Each week, the highest-ranked stocks form the long sleeve and the lowest-ranked stocks form the short sleeve. Headline gross signal comparisons use the same equal-weight high-minus-low portfolio construction across models, while alternative weighting and portfolio-construction methods are examined as robustness tests in the appendix. For implementation-aware tests, prior holdings are carried forward from one rebalance to the next, so trading costs depend on the changes required to reach the new portfolio. Rebalancing accounts for stock-specific spreads, forecast liquidity and volatility, nonlinear market impact, and ADV-based position and trade limits. The transaction-cost model, portfolio-construction variants, and execution-input forecasting layer are described in Appendix F.

3 Models and Implementation

The CNN and RF produce separate probability forecasts. The 50/50 rank blend averages their within-date cross-sectional ranks. The logistic stack estimates a single walk-forward mapping from the standardized CNN and RF logits to the five-day directional outcome and therefore provides a fixed-mapping benchmark for the HMM. The expected-return fusion instead maps the same expert logits directly into an expected five-day return, which is used as a cross-sectional ranking signal. The HMM allows the mapping from the expert signals to the final forecast to vary with a latent weekly state. State probabilities evolve through the estimated Markov transition process, while the state-specific forecasting equations use the expert signals and date-level context. Its final forecast averages the state-specific predictions using the one-step-ahead state distribution, so current outcomes are not used to form

the current forecast. Mathematical definitions of the fusion benchmarks and the complete HMM factorization, state-specific emissions, filtering equations, and tuning protocol are reported in Appendix B and Appendix E.

The implementation analysis asks whether differences in predictive performance survive realistic trading frictions and increasing scale. Each rebalance begins from the portfolio carried from the previous week, so costs depend on the trades required to move toward the new target weights. Portfolio construction penalizes risk, spread crossing, and nonlinear market impact, while limiting both position size and weekly trading relative to forecast ADV. We evaluate fixed AUM from \$10 million to \$1 billion to isolate the effect of scale from portfolio compounding; dynamic-NAV results are reported separately as a supplementary stress test. The full cost model, portfolio constraints, and numerical implementation are described in Appendix F.

4 Results

4.1 Gross Performance and Signal Persistence

Table 1 reports the common-universe equal-weight results. The central finding is that fusion works, but complexity does not monotonically improve gross performance. The untuned 50/50 rank blend has the highest Sharpe, 4.957, followed by the logistic stack, expected-return fusion, and HMM. HMM nevertheless improves modestly on RF and substantially outperforms CNN. A paired block bootstrap gives an HMM-minus-RF Sharpe difference of +0.319 with a 95% interval of approximately $[+0.191, +0.453]$. Against the rank blend, however, HMM’s Sharpe difference is -0.249 , with a 95% interval of $[-0.450, -0.049]$. Comparisons with the remaining fusion methods are reported in Appendix Table 4.

Table 1: Common-universe equal-weight gross performance, 2001–2024. Turnover measures how much the model’s desired long and short portfolios change from one week to the next, averaged across weekly rebalances.

Model	Ann. return	Ann. vol.	Sharpe	Turnover
CNN	14.36%	11.99%	1.198	1.794
RF	107.64%	24.52%	4.389	0.712
Rank blend 50/50	91.23%	18.40%	4.957	1.534
Logistic stack	101.97%	21.50%	4.743	1.271
HMM	107.89%	22.92%	4.708	1.073
Expected-return fusion	105.88%	22.41%	4.725	1.157

Spanning regressions sharpen that distinction. HMM has a significant annualized alpha of 11.77% relative to RF alone and 3.74% relative to RF and CNN jointly, indicating that its return stream is not fully spanned by the two standalone expert strategies. HMM alpha is not statistically distinguishable from zero, however, relative to the 50/50 rank blend, logistic stack, or expected-return fusion (Appendix Table 5). The strongest evidence is therefore for the value of combining complementary signals, rather than for a unique latent-state source of gross alpha.

The fusion methods also differ sharply in persistence. RF has rank persistence of about 0.81 and the lowest target turnover; HMM is next at about 0.65, followed by expected-return fusion at 0.60, logistic stacking at 0.53, and the rank blend at 0.36 (Appendix Table 6). HMM therefore sacrifices some RF stability to incorporate complementary CNN information, but filters substantially more week-to-week variation than the simpler fusion rules. The difference in signal stability becomes central once transaction costs and capacity are introduced.

4.2 Implementation and capacity

Figure 1 shows how the model ranking changes once transaction costs and scale are introduced across fixed AUM levels from \$10 million to \$1 billion. HMM has the highest net Sharpe at \$10 million and \$25 million, with values of 3.850 and 3.414, respectively. RF narrowly overtakes HMM at \$50 million, and its advantage widens with scale: at \$100 million, RF has a net Sharpe of 2.939 versus 2.628 for HMM, while at \$1 billion the corresponding values are 1.726 and 1.396. At \$100 million, paired block-bootstrap inference gives an HMM-minus-RF Sharpe difference of -0.311 , with a 95% interval of $[-0.508, -0.121]$. The full fixed-AUM results and additional bootstrap comparisons are reported in Appendix Table 7 and Table 8.

The crossover is not driven by a reversal in pre-cost performance. At \$1 billion, HMM still has a higher executed gross Sharpe than RF, approximately 2.85 versus 2.70, but its higher trading costs reduce its net Sharpe below that

of RF. The change in ranking is therefore an implementation effect rather than a disappearance of HMM's gross performance advantage (Appendix G.2).

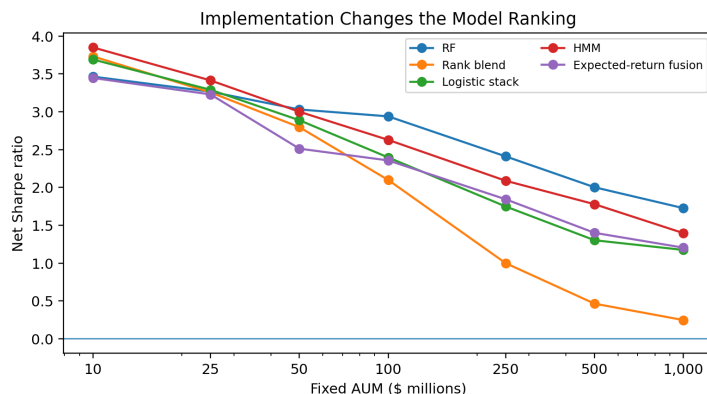


Figure 1: Net Sharpe across fixed AUM from \$10 million to \$1 billion. HMM leads at low AUM, while RF's Sharpe becomes superior as scale increases. CNN is omitted because its after-cost Sharpe is negative throughout the tested AUM grid.

4.3 Why the ranking reverses

At \$100 million, HMM incurs approximately 76.4 bps of trading cost per weekly rebalance, compared with 47.5 bps for RF. Trading costs consist of a spread-crossing component, modeled as proportional to the amount traded (R. Almgren and Chriss 2001), and a nonlinear market-impact component whose total cost increases more than proportionally with trade size (R. F. Almgren 2003; R. Almgren, Thum, et al. 2005). Nearly the entire RF–HMM cost gap is attributable to HMM's greater turnover, and about 90% of the incremental cost comes from nonlinear impact. Similar spread, ADV, and volatility characteristics across the two portfolios indicate that HMM is more expensive primarily because it trades more, not because it systematically trades more costly securities (Appendix Table 9).

The short sleeve is the main bottleneck. At \$100 million, HMM's short-side trading cost per unit of turnover is roughly twice its long-side value, at approximately 102 versus 50 bps. Short-sleeve stocks have lower ADV, wider spreads, and higher volatility, and roughly two-thirds of HMM's incremental cost relative to RF is concentrated there (Appendix Table 9).

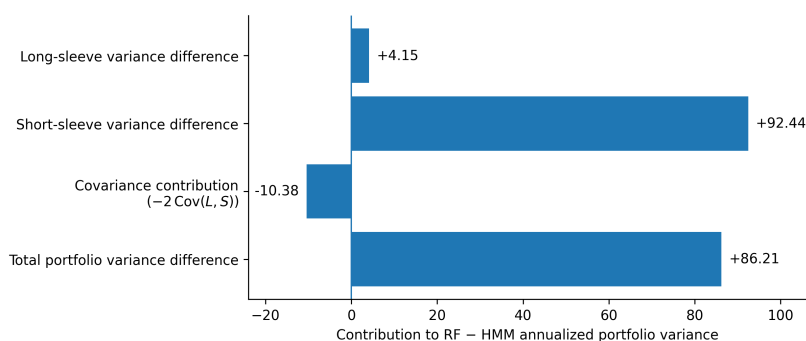


Figure 2: RF–HMM portfolio variance decomposition at \$100 million. The bars show how differences in long-sleeve variance, short-sleeve variance, and long–short covariance contribute to the total difference in annualized long–short portfolio variance. Positive bars increase RF's variance relative to HMM. Exact component values are reported in Appendix Table 11.

At \$100 million, the implemented RF portfolio has annualized volatility of 26.33%, compared with 24.64% for HMM. Figure 2 decomposes the corresponding RF–HMM variance difference. In annualized variance units scaled by 10^4 , the short sleeve contributes +92.44, the long sleeve +4.15, and the covariance term -10.38 . The same

pattern holds across the AUM grid (Appendix G.4, Table 10). Taken together, the cost and variance decompositions identify the short sleeve as the central RF–HMM trade-off: it accounts for most of HMM’s risk advantage, but it is also where HMM’s additional turnover is most costly to execute.

Diagnostic sleeve swaps support the same interpretation. At \$100 million, a portfolio using HMM’s long sleeve and RF’s short sleeve has Sharpe 2.967, close to RF’s 2.939, while the opposite combination has Sharpe 2.609, close to HMM’s 2.628; neither difference is statistically significant. At \$1 billion the matching is nearly exact. Because these portfolios are not jointly reoptimized, they are diagnostic rather than implementable alternatives, and they show that the short sleeve carries much of the RF–HMM after-cost performance distinction at moderate and high AUM (Appendix G.6, Table 13).

4.4 What the latent states do

Direct mechanism tests do not show that changes in the latent state predict higher HMM returns relative to the fixed-mapping logistic stack. They are, however, strongly associated with portfolio changes. When the dominant HMM state changes, target turnover rises by about 0.38. Target turnover measures changes in the desired long and short portfolio weights before transaction costs or trading constraints are applied; a value of 1 corresponds to turnover equal to the full notional of one sleeve. A post-hoc attribution proxy, constructed from cross-sectional regressions of HMM forecast logits on standardized CNN and RF logits, shows that changes in the relative CNN–RF attribution mix account for about 22% of the week-to-week variation in target turnover (Appendix G.5, Table 12).

The state-transition relationship persists after implementation. At \$100 million, dominant-state changes are associated with 0.29 higher realized turnover—equivalent to an additional 29% of one sleeve’s notional—and a 12.5 percentage-point increase in the average fraction of stocks replaced across the long and short sleeves (Appendix Table 12).

The evidence indicates that the latent states primarily govern how the CNN and RF signals are combined and translated into portfolio positions, rather than identifying periods of systematically higher incremental HMM returns (Appendix G.5). The state-specific mappings differ in how they use the expert signals, date-level context, and CNN–RF disagreement, allowing the fusion rule to adapt over time (Appendix G.9). At the same time, HMM rankings are more persistent than those of the other fusion methods (Appendix Table 6), so the portfolio typically changes less from week to week; weeks with a dominant-state transition have about 0.38 higher target turnover than weeks without a transition (Appendix G.5). The HMM is therefore more appropriately interpreted as a persistent, state-dependent fusion mechanism than as a conventional market-regime classifier.

5 Discussion, Limitations, and Conclusion

CNN and RF contain complementary return information, and fusion produces economically meaningful gains relative to the standalone experts. HMM has significant alpha relative to RF and to RF and CNN jointly, but not relative to the simpler fusion benchmarks. Higher-gross-Sharpe fusion rules generally produce less persistent rankings and greater turnover. RF is highly persistent and less costly to implement but more sector concentrated and higher risk, while HMM occupies an intermediate position, retaining much of fusion’s diversification benefit with greater stability than the simpler fusion rules.

That trade-off explains the AUM-dependent ranking. At low AUM, HMM’s lower risk is valuable enough to offset its higher trading cost. As AUM rises, nonlinear impact increasingly penalizes its residual turnover, particularly in the structurally expensive short sleeve, and RF’s persistence becomes more valuable.

Several limitations remain. The execution model is simulated rather than validated against realized institutional fills, and the strongest mechanism diagnostics were developed after observing the main results and therefore warrant confirmation on untouched data. Formal optimizer convergence flags are frequent under the production stopping rule, although higher-iteration re-solves produce effectively identical objectives and weights; a KKT-style diagnostic would provide a stronger convergence check (Appendix Table 14).

The main conclusion is twofold: multimodal fusion adds economically meaningful return information, but the best gross ensemble need not be the best implementable portfolio. The relevant trade-off is between diversification and persistence: HMM is strongest at low AUM, while RF scales more efficiently because its signal requires less trading. The short sleeve drives both sides of this trade-off, providing most of HMM’s risk reduction while imposing its largest implementation burden. More generally, ignoring existing holdings, turnover, nonlinear impact, and capacity can materially misstate which forecasting improvements are investable.

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A Extended related work

A.1 Chart-based, structured-context, and hybrid prediction

Whether past prices contain predictive information about future returns is a longstanding question in finance. Lo, Mamaysky, and Wang (2000) gave technical analysis a more formal empirical treatment, while Jiang, Kelly, and Xiu (2023) replaced pre-specified chart rules with learned visual patterns by representing price and volume histories as images and using a CNN to predict returns. Together, they motivate the visual branch of our design and show that image-based price prediction can be treated as a forecasting approach rather than as a purely heuristic trading rule.

A separate line of work studies return prediction using structured financial information observed at the forecast date. Bastianello (2026), for example, shows how rich financial information can be assembled into an ex-ante machine-learning benchmark under strict timing discipline. Although that paper studies analysts' beliefs rather than multimodal stock prediction, it is useful here because it illustrates how accounting, market, and macroeconomic information can be converted into a real-time forecasting signal without violating the information set available at the prediction date. Our structured-context branch follows that broader point-in-time logic.

Hybrid prediction models increasingly combine multiple sources of financial information, but typically in ways different from our design. Picasso et al. (2019) combine technical signals with news sentiment, while T. Kim and H. Y. Kim (2019) fuse a stock-chart CNN with an LSTM built from price-and-volume sequences derived from the same underlying market stream. By contrast, our setting combines two economically distinct branches: a chart-based signal and a separate structured-context signal. The two branches are trained separately and are fused only after each branch has produced an out-of-sample probability forecast, which makes the paper closer to a late-fusion multimodal design than to end-to-end feature fusion within a single market-data stream.

B Detailed predictive formulation

B.1 Prediction problem and notation

Let i index stocks and t index weekly decision dates. The main prediction target is

$$Y_{i,t} \equiv Y_{i,t}^{(5)} = \mathbb{1}\{R_{i,t}^{(5)} > 0\},$$

where $R_{i,t}^{(5)}$ denotes the forward 5-trading-day return. For the multihorizon HMM panel, we also define

$$Y_{i,t}^{(20)} = \mathbb{1}\{R_{i,t}^{(20)} > 0\},$$

where $R_{i,t}^{(20)}$ is the forward 20-trading-day return.

Let $X_{i,t}^{img}$ denote the chart-image input and $X_{i,t}^{tab}$ the tabular input, and let $\mathcal{D}_{1:t}$ denote the observed data available through week t . The latent weekly state is denoted by

$$S_t \in \{1, \dots, K\}.$$

Our objective is to estimate the probability that the next 5-day return is positive using a late-fusion model that combines chart-based and tabular probability forecasts in a state-dependent way.

B.2 Base expert forecasts

The CNN and RF are treated as probability-forecasting experts. The CNN produces the short-horizon chart signal

$$p_{i,t}^{cnn} = P(Y_{i,t}^{(5)} = 1 \mid X_{i,t}^{img}),$$

and the RF produces the medium-horizon structured-context signal

$$p_{i,t}^{rf} = P(Y_{i,t}^{(20)} = 1 \mid X_{i,t}^{tab}).$$

The final evaluation target remains the 5-day direction; the RF probability is used as a medium-horizon structured-context expert within the multihorizon HMM. Define the expert log-odds as

$$\ell_{i,t}^{cnn} = \text{logit}_\varepsilon(p_{i,t}^{cnn}), \quad \ell_{i,t}^{rf} = \text{logit}_\varepsilon(p_{i,t}^{rf}),$$

where, for numerical stability,

$$\text{logit}_\varepsilon(p) = \log\left(\frac{\tilde{p}}{1 - \tilde{p}}\right), \quad \tilde{p} = \min\{1 - \varepsilon, \max\{\varepsilon, p\}\}.$$

For notational simplicity, we write $\ell_{i,t}^m = \text{logit}_\varepsilon(p_{i,t}^m)$ for $m \in \{cnn, rf\}$ below.

The HMM uses these expert logits directly. The logistic and expected-return benchmarks instead standardize each expert logit using the mean and standard deviation estimated from the corresponding historical training sample. The 50/50 rank blend instead operates on the within-date cross-sectional ranks of the original expert probabilities. Construction and training details for the two base branches are reported in Appendix C.1.

B.3 Non-state fusion benchmarks

The three non-HMM fusion benchmarks combine the same CNN and RF expert forecasts without introducing a latent state.

For the 50/50 rank blend, let $\mathcal{R}_t(\cdot)$ denote the within-date cross-sectional percentile-rank transformation and define

$$r_{i,t}^{cnn} = \mathcal{R}_t(p_{i,t}^{cnn}), \quad r_{i,t}^{rf} = \mathcal{R}_t(p_{i,t}^{rf}).$$

The combined score is

$$s_{i,t}^{50/50} = \frac{1}{2}r_{i,t}^{cnn} + \frac{1}{2}r_{i,t}^{rf}.$$

The rank transformation places the two expert signals on a common cross-sectional scale despite differences in their probability distributions and forecast horizons. The equal-weight average therefore gives each expert the same weight in the combined score without estimating combination weights or requiring the two probability forecasts to be calibrated on the same scale. The resulting $s_{i,t}^{50/50}$ is therefore a cross-sectional ranking score rather than a calibrated probability. Because the combination weights are fixed rather than estimated, the 50/50 blend is best interpreted as an untuned statistical benchmark rather than as a structural or economically motivated fusion rule. Its purpose is to test whether simple diversification across the two expert rankings is sufficient to capture their complementary information.

The logistic and expected-return benchmarks instead operate on standardized expert logits. For prediction year y , define

$$\tilde{\ell}_{i,t}^m = \frac{\ell_{i,t}^m - \bar{\ell}_y^m}{s_y^m}, \quad m \in \{cnn, rf\},$$

where $\bar{\ell}_y^m$ and s_y^m are respectively the training-sample mean and standard deviation of expert m 's logit, estimated using only the historical sample available for the year- y fit.

The walk-forward logistic stack maps these standardized logits into a 5-day directional probability:

$$\hat{p}_{i,t}^{log} = \sigma\left(\alpha_y + \beta_{cnn,y}\tilde{\ell}_{i,t}^{cnn} + \beta_{rf,y}\tilde{\ell}_{i,t}^{rf}\right),$$

where $\sigma(x) = 1/(1 + e^{-x})$. The coefficients are estimated on completed historical 5-day directional labels by minimizing the L_2 -regularized Bernoulli negative log-likelihood,

$$\min_{\alpha, \beta} - \sum_{(i,t) \in \mathcal{T}_y} \left[Y_{i,t}^{(5)} \log \hat{p}_{i,t}^{log} + (1 - Y_{i,t}^{(5)}) \log (1 - \hat{p}_{i,t}^{log}) \right] + \lambda_{\log} \|\beta\|_2^2,$$

where $\mathcal{T}_y$ contains only observations whose 5-day outcomes are fully realized before prediction year y . On the weekly grid, the logistic and expected-return benchmarks therefore omit the most recent historical weekly observation from each annual fit. The multihorizon HMM uses a four-week label lag because its intermediate 20-day outcome must also be fully realized before entering the training sample. Because $Y_{i,t}^{(5)}$ is binary, the likelihood criterion directly estimates the conditional probability of a positive 5-day return, while the L_2 penalty shrinks the expert coefficients toward zero. The logistic stack therefore serves as the fixed-mapping probability benchmark for the HMM.

The expected-return fusion asks a different question. Rather than estimating the probability of a positive return, it models the realized 5-day return directly as

$$R_{i,t}^{(5)} = \alpha_y^{ER} + \beta_{cnn,y}^{ER} \tilde{\ell}_{i,t}^{cnn} + \beta_{rf,y}^{ER} \tilde{\ell}_{i,t}^{rf} + \varepsilon_{i,t}.$$

Its prediction is therefore

$$\hat{\mu}_{i,t}^{ER} = \hat{E}\left[R_{i,t}^{(5)} \mid \tilde{\ell}_{i,t}^{cnn}, \tilde{\ell}_{i,t}^{rf}\right] = \alpha_y^{ER} + \beta_{cnn,y}^{ER} \tilde{\ell}_{i,t}^{cnn} + \beta_{rf,y}^{ER} \tilde{\ell}_{i,t}^{rf}.$$

The coefficients are estimated by walk-forward Ridge regression,

$$\min_{\alpha, \beta} \sum_{(i,t) \in \mathcal{T}_y} \left(R_{i,t}^{(5)} - \alpha - \beta^\top \tilde{\ell}_{i,t} \right)^2 + \lambda_{ER} \|\beta\|_2^2,$$

where

$$\tilde{\ell}_{i,t} = \begin{pmatrix} \tilde{\ell}_{i,t}^{cnn} \\ \tilde{\ell}_{i,t}^{rf} \end{pmatrix}.$$

Under squared-error loss, the population-optimal predictor is the conditional mean. The linear Ridge specification therefore estimates a regularized linear approximation to the conditional expected 5-day return given the two expert signals. The Ridge penalty stabilizes the estimated combination when the two expert signals contain correlated information. The resulting $\hat{\mu}_{i,t}^{ER}$ is used as a cross-sectional ranking signal rather than as a calibrated probability.

B.4 HMM inputs

Using the expert logits defined above, the HMM input vector is

$$Z_{i,t} = \left(\ell_{i,t}^{cnn}, \ell_{i,t}^{rf}, c_t \right),$$

where c_t contains date-level context variables shared across stocks at week t . Construction of the multihorizon HMM panel and the date-level context variables is described in Appendix E.

B.5 Hidden Markov state structure

The fusion layer is a hidden Markov model with latent weekly state S_t . We assume first-order Markov dynamics:

$$P(S_t = k \mid S_{t-1}, S_{t-2}, \dots) = P(S_t = k \mid S_{t-1}),$$

with transition probabilities

$$a_{jk} = P(S_t = k \mid S_{t-1} = j), \quad \sum_{k=1}^K a_{jk} = 1.$$

The HMM allows the mapping from expert signals to outcome probabilities to vary across latent states. The final 5-day forecast is therefore obtained from a state-mixture model rather than from a fixed linear blend.

B.6 Joint probability factorization

Given the hidden state S_t and inputs $Z_{i,t}$, the HMM models the joint state-conditional distribution

$$P\left(Y_{i,t}^{(5)}, Y_{i,t}^{(20)} \mid Z_{i,t}, S_t\right).$$

Using the chain rule of probability,

$$P\left(Y_{i,t}^{(5)}, Y_{i,t}^{(20)} \mid Z_{i,t}, S_t\right) = P\left(Y_{i,t}^{(5)} \mid Y_{i,t}^{(20)}, Z_{i,t}, S_t\right) P\left(Y_{i,t}^{(20)} \mid Z_{i,t}, S_t\right).$$

The two-block structure allows the 20-day direction to capture a broader directional component, while the 5-day block refines the short-horizon forecast conditional on it.

B.7 State-specific logistic emissions

Using the sigmoid function defined above, conditional on $S_t = k$, the 20-day block is

$$P\left(Y_{i,t}^{(20)} = 1 \mid Z_{i,t}, S_t = k\right) = \sigma\left(\alpha_k^{(20)} + \beta_{k,rf}^{(20)} \ell_{i,t}^{rf} + \gamma_k^{(20)\top} c_t\right),$$

and the 5-day block is

$$P\left(Y_{i,t}^{(5)} = 1 \mid Y_{i,t}^{(20)}, Z_{i,t}, S_t = k\right) = \sigma\left(\alpha_k^{(5)} + \beta_{k,cnn}^{(5)} \ell_{i,t}^{cnn} + \beta_{k,rf}^{(5)} \ell_{i,t}^{rf} + \delta_k Y_{i,t}^{(20)} + \gamma_k^{(5)\top} c_t\right).$$

Here $Y_{i,t}^{(20)}$ denotes the intermediate 20-day binary outcome rather than a probability forecast. Within each historical refit block, the emission models are estimated using an EM-style routine. In the training sample, both $Y_{i,t}^{(20)}$ and $Y_{i,t}^{(5)}$ are observed, so the 20-day and 5-day logistic blocks are fit on observed labels using state-posterior weights. At prediction time, by contrast, $Y_{i,t}^{(20)}$ is unobserved, so the final 5-day forecast is obtained by marginalizing over the model-implied distribution of that intermediate 20-day outcome. Additional implementation details are reported in Appendix E.

B.8 Bayesian filtering and state updating

Let $\pi_{t|t-1}(k) = P(S_t = k \mid \mathcal{D}_{1:t-1})$ and $\pi_{t|t}(k) = P(S_t = k \mid \mathcal{D}_{1:t})$ denote the one-step-ahead and filtered probabilities of state k , respectively.

Define the transition probabilities by $a_{jk} = P(S_t = k \mid S_{t-1} = j)$, and let $L_t(k) = P(\mathcal{D}_t \mid S_t = k)$ denote the week- t likelihood under state k .

The prediction step is

$$\pi_{t|t-1}(k) = \sum_{j=1}^K a_{jk} \pi_{t-1|t-1}(j),$$

and the Bayesian update step is

$$\pi_{t|t}(k) = \frac{L_t(k) \pi_{t|t-1}(k)}{\sum_{m=1}^K L_t(m) \pi_{t|t-1}(m)}.$$

At prediction time, we use $\pi_{t|t-1}(k)$, since week- t outcomes are not yet observed.

B.9 Final predictive probability and walk-forward estimation

The HMM forecast for stock i at week t is obtained by marginalizing over the predicted latent-state distribution:

$$\hat{p}_{i,t}^{hmm} = P\left(Y_{i,t}^{(5)} = 1 \mid Z_{i,t}, \mathcal{D}_{1:t-1}\right) = \sum_{k=1}^K \pi_{t|t-1}(k) P\left(Y_{i,t}^{(5)} = 1 \mid Z_{i,t}, S_t = k\right).$$

Within state k , this state-specific 5-day success probability is obtained by marginalizing over the unobserved intermediate 20-day outcome:

$$P(Y_{i,t}^{(5)} = 1 \mid Z_{i,t}, S_t = k) = \sum_{y \in \{0,1\}} P(Y_{i,t}^{(5)} = 1, Y_{i,t}^{(20)} = y \mid Z_{i,t}, S_t = k).$$

Using the chain rule,

$$P(Y_{i,t}^{(5)} = 1 \mid Z_{i,t}, S_t = k) = \sum_{y \in \{0,1\}} P(Y_{i,t}^{(5)} = 1 \mid Y_{i,t}^{(20)} = y, Z_{i,t}, S_t = k) P(Y_{i,t}^{(20)} = y \mid Z_{i,t}, S_t = k).$$

Substituting this decomposition into the state mixture gives

$$\hat{p}_{i,t}^{hmm} = \sum_{k=1}^K \pi_{t|t-1}(k) \sum_{y \in \{0,1\}} P(Y_{i,t}^{(5)} = 1 \mid Y_{i,t}^{(20)} = y, Z_{i,t}, S_t = k) P(Y_{i,t}^{(20)} = y \mid Z_{i,t}, S_t = k).$$

Estimation is fully walk-forward. Predictions at date t use only $\mathcal{D}_{1:t-1}$ together with current inputs $Z_{i,t}$. Hyperparameters for prediction year y are selected by an annual expanding-window procedure using only data available through year $y - 1$. After selection, the chosen specification is run in a walk-forward manner over year y , and only the resulting year- y predictions are retained for evaluation. Additional implementation details are reported in Appendix E.

C Supplementary material

C.1 Data construction and preprocessing

C.1.1 Visual branch construction

The visual branch uses daily OHLCV data to generate stock-chart images for a plain CNN. While the full generator can produce 5-, 20-, or 60-period images, the main specification uses the short-horizon setting: 5-day lookback images paired with a 5-day prediction target. Images are stored as uint8 memory-mapped arrays, with labels and metadata recorded in Feather format.

Image construction follows the Jiang-style normalization logic. For each stock and period-end date, a trailing OHLC window with moving averages is formed, and prices are rescaled so that the first closing price is normalized to 1. Subsequent closes are reconstructed from returns, and the daily open, high, and low are expressed relative to the adjusted close. In the specification used here, the final CNN inputs are standard bar-chart images with both volume bars and moving-average overlays. For 5-day images, the stored chart size is 15×32 pixels. When necessary, period-end dates are inferred directly from available trading dates.



Figure 3: Example 5-day stock-chart image used as input to the CNN. The chart uses the project’s short-horizon specification with volume bars and moving-average overlays.

The CNN dataset wrapper loads the saved image arrays and label files, resolves the relevant forward-return column, and converts the task to binary classification by assigning a label of 1 when the forward return is positive and 0 otherwise. Observations with missing labels are removed. The wrapper also standardizes the images using an in-sample mean and standard deviation saved to disk; for the 2D image specification used here, these moments are computed from the first 50,000 images in the relevant in-sample dataset.

Model training follows the standard image-CNN path rather than the alternative time-series branch. The core specification uses a 5-day window and a 5-day prediction horizon, with bar charts, moving averages, volume bars, and a 70/30 random split between training and validation within the in-sample years. The final forecasts are produced by training an ensemble of CNN runs and averaging their predicted up-probabilities out-of-sample.

C.1.2 Tabular branch construction

The tabular branch is built from the merged weekly-with-ratios panel used by the random-forest pipeline. The dataset utilities enforce a strict as-of design, meaning that each stock-date observation is constructed using only information available at that date. Dates are standardized, keys and targets are converted to numeric form, and optional staleness filtering is applied via `ratio_age_days` to limit reliance on outdated accounting information. Rows without matched ratio data are preserved by default and are removed only if later preprocessing steps require it. Feature columns are then inferred by excluding reserved metadata and target columns and, where specified in the configuration, adding a small set of extra structured predictors.

A central design choice is that `sector_id` is treated as a Fama–French 49 industry bucket for preprocessing rather than as a forecasting feature. Before model fitting, sector IDs are validated to ensure they are present and fall within the admissible 1–49 range. Missing feature values are then imputed only within each $(Date, sector_id)$ cross-section, closely matching the intended industry-median logic. Rows that still cannot be resolved after this strict FF49-by-date imputation step are dropped rather than sent through broader fallback rules. The binary RF label is constructed from the sign of the forward 20-trading-day return.

C.1.3 Common stock-date overlap

All model comparisons are restricted to a common stock-date universe. The overlap utilities construct a merged RF–CNN panel, identify the $(Date, StockID)$ keys on which both expert probabilities are available, and extract year-specific overlap keys from that shared frame. Individual signal files are then restricted to the same keys as before portfolio construction, so that the CNN, RF, and ensemble models are evaluated on the same underlying sample.

C.1.4 Evaluation metric choice

The main evaluation uses ranked probabilities rather than a fixed classification threshold. The ranking-based evaluation follows the economic use case: the portfolio rule sorts stocks by predicted up-probability and trades the top and bottom deciles, so the relevant object is the cross-sectional ranking induced by each model. Threshold-dependent metrics such as precision and recall are therefore less central to the reported comparison. A model can have modest threshold classification performance but still be economically useful if it ranks stocks well enough to separate the top and bottom deciles after common-universe restrictions, tradability screens, and transaction-cost adjustments.

D Secondary preprocessing and metadata repair

D.1 Sector-metadata repair

In standard finance usage, the FF49 industry classification is typically treated as a deterministic mapping from SIC codes rather than as a learned object (Kenneth R. French Data Library 2026). In our setting, missing or invalid FF49 labels therefore create an upstream data-quality problem for preprocessing rather than a standalone prediction task. The sector-recovery stage is best understood as a conservative metadata-repair module that recovers only sufficiently confident FF49 labels, enabling stricter, more consistent within-industry preprocessing without changing the downstream random-forest model.

A secondary but important issue in the tabular branch is the presence of missing or invalid FF49 sector identifiers. The strict random-forest pipeline treats `sector_id` as required metadata for within-industry imputation rather than as a forecasting feature. In the baseline RF dataset code, `sector_id` must exist, must lie in the admissible 1–49 range, and is then used to impute missing predictors only within each $(Date, sector_id)$ cross-section. Rows that remain unresolved after this strict FF49-by-date imputation step are dropped rather than passed through broader fallback rules.

We address the resulting data loss with a separate sector-recovery pipeline. The design is intentionally conservative. It treats sector recovery as metadata repair rather than as the main forecasting task, runs in a walk-forward expanding-window fashion, allows low-confidence predictions to be rejected, and can drop still-unresolved rows before the patched parquet is handed back to the RF pipeline. Burn-in requirements are also explicit: the repair model is not allowed to activate until there are enough historically labeled rows and enough broad and fine industry classes are present.

The sector-recovery stage is conceptually closer to structured classification with abstention than to standard asset-pricing practice. Supervised sector classification from financial statements provides the closest broad analogue (Heijden 2022), while the confidence-thresholded acceptance rule follows the logic of selective classification with abstention, under which uncertain cases are excluded from automatic prediction rather than forced into a label (Geifman and El-Yaniv 2017).

The repair model itself is a supervised multinomial logistic classifier. In its preferred form, it is hierarchical: it first predicts a broad FF12-style sector group, then predicts the FF49 class conditional on that broad-group assignment. Median imputation and optional missingness indicators are used only within this classifier; they are local to the metadata-repair step and do not alter the downstream strict RF feature-construction logic. When no valid FF49-to-FF12 mapping is supplied, the code falls back to a direct FF49 multinomial logistic model.

D.2 Walk-forward repair protocol and threshold selection

The repair pass is run annually. For each prediction year, the engine reads only the minimal columns needed for sector recovery, loads the historical training slice through year $t - 1$, and then predicts missing or invalid sector IDs for year t . If burn-in conditions are not yet met, the engine records this and declines to recover sectors for that year rather than forcing noisy labels.

Thresholds for accepting recovered sector IDs are determined through a masked-label walk-forward audit. Rows with known sector labels are temporarily hidden within a prediction year, predicted out-of-sample using only information available at the time, and then compared with the truth. The threshold grid is evaluated by coverage and accepted-set precision rather than unconditional accuracy alone. In the hierarchical case, acceptance depends on both the broad-group confidence τ_{12} and the within-group confidence τ_{49} ; the default configuration targets high accepted-set precision and stores frozen thresholds once selected.

Table 2 reports the annual acceptance counts and acceptance rates from the walk-forward sector-recovery procedure. Acceptance is negligible in the earliest years and rises materially only once the repair model has accumulated enough labeled history to operate more effectively.

Table 2: Walk-forward sector-recovery acceptance by prediction year

Prediction year	Rows	Accepted rows	Accepted rate
2008	439	24	0.054670
2009	1389	118	0.084953
2010	3742	354	0.094602
2011	6058	906	0.149554
2012	9210	1920	0.208469
2013	12575	2886	0.229503
2014	18126	3988	0.220015
2015	23424	5044	0.215335
2016	24148	5236	0.216830
2017	27794	5956	0.214291
2018	34371	7250	0.210934
2019	39262	7802	0.198716
2020	48423	8951	0.184850
2021	63635	9293	0.146036
2022	74000	11196	0.151297
2023	72456	11947	0.164886
2024	69894	12699	0.181689

At prediction time, the hierarchical classifier returns both broad-group and fine-group probabilities. A recovered FF49 label is accepted only when the broad-group probability exceeds τ_{12} , the conditional FF49 probability exceeds τ_{49} , and a fine label is available. In direct FF49 mode, acceptance depends only on the FF49 probability threshold. Cases that do not meet those confidence requirements remain unresolved rather than being forced into the forecasting panel (Geifman and El-Yaniv 2017).

D.3 Patching repaired sectors back into the RF parquet

Recovered sector IDs are patched back into the original weekly-with-ratios parquet rather than fed into the RF trainer through a separate side channel. The patcher merges walk-forward recovery predictions back onto the

original $(StockID, Date)$ keys, preserves the original sector values for audit purposes, adds recovered and final sector columns, and records whether a repaired label was actually used. If configured to do so, it then drops rows whose final sector ID remains invalid. The resulting output is a walk-forward-safe, patched parquet file that the strict RF pipeline can consume without any changes to its own preprocessing logic.

D.4 Additional RF data checks

Outside the sector-recovery module, the tabular pipeline also includes hard checks on the weekly-with-ratios panel. These checks verify the uniqueness of $(StockID, Date)$ keys, enforce $decision_date \leq Date$, require $ratio_date \leq decision_date$ when a ratio date exists, and verify that $ratio_age_days$ matches the implied difference between $decision_date$ and $ratio_date$. Their purpose is not to improve fit directly, but to enforce a defensible as-of design before feature construction begins.

E Ensemble panel construction and HMM estimation

E.1 Ensemble panel and common universe

The ensemble layer is built from saved CNN and RF prediction files rather than by refitting the base models within the portfolio code. Both sources are first standardized onto common Date and StockID keys, after which the expert probabilities are renamed p_cnn and p_rf and merged into a unified expert panel. When tuning requires realized outcomes, the panel is further augmented with a binary target y and a raw forward-return column fwd_ret from the period-return table.

The ensemble panel is restricted to the common stock-date overlap universe described in the main text, ensuring that RF, CNN, and HMM signals are evaluated on the same underlying rows.

E.2 HMM modeling panel

The HMM is estimated on a richer multihorizon panel built by `HMMDataBuilder`, rather than the simple two-expert panel used for fixed blends. The panel starts with the merged expert frame, treats the CNN as the 5-day expert and the RF as the 20-day expert, and augments those signals with realized 5-day and 20-day forward returns, along with the corresponding binary labels y_5d and y_20d .

The builder also enriches the panel with stock-level context features drawn from an external feature panel. Once aggregated to the date level, these features are merged back onto each stock-date row, along with cross-sectional summaries of the expert panel, including the means of p_cnn and p_rf and the moments of the absolute expert gap. The HMM then uses these variables as date-level context features, allowing the fusion rule to adapt not only to the expert probabilities themselves but also to broader changes in the cross-sectional forecasting environment.

E.3 HMM estimation and prediction

The HMM is a date-level latent-state model with state-specific logistic emission blocks. The 20-day block is fit on the RF expert and date-level context, while the 5-day block is fit on both the CNN and RF experts, the intermediate 20-day outcome, and the same contextual information.

Within each historical refit block, estimation proceeds by an EM-style routine. The code first initializes latent states using K-means on a compact set of date-level context features and initializes the transition matrix from that hard state path. It then fits state-specific weighted logistic regressions for the 20-day and 5-day emission models. Given those emission models, the code runs the forward-backward algorithm to compute posterior state probabilities and expected transition counts. It next updates the initial-state probabilities and transition matrix, refits the weighted logistic emissions using the posterior state weights, and repeats until the absolute change in log-likelihood falls below the convergence threshold or the iteration cap is reached.

Prediction is fully walk-forward. For each eligible prediction block, the model is fit only on the historical window allowed by the label-lag rule, the filtered state distribution at the end of that training block is propagated forward through the transition matrix, and final stock-level 5-day probabilities are obtained by averaging over the resulting state distribution. The code stores both the row-level forecasts and the date-level predicted state probabilities for later diagnostics. Additional interpretation of the resulting state dynamics is reported in Appendix G.5 and Appendix G.9.

E.4 Annual tuning protocol

Hyperparameters are selected by a separate annual expanding-window tuner. For prediction year y , the tuner uses only data through $y - 1$, scores candidate HMM specifications on recent completed validation years, and freezes the chosen configuration before the year- y run. The search space is deliberately narrow, covering the main HMM controls: number of states, L2 regularization, refit frequency, training-window length, and transition smoothing.

The annual runner then fits the selected HMM specification on the available historical sample and retains only the target year’s walk-forward predictions. The HMM is handled outside the generic ensemble runner because it requires the multihorizon panel and a dedicated estimation loop. Keeping that workflow separate preserves a transparent tuning protocol while allowing the latent-state model to adapt within each historical fit.

F Transaction-cost forecasting and implementation realism

F.1 Purpose

The transaction-cost forecasting layer plays a supporting rather than a central role in the project. Its purpose is to improve implementation realism by supplying execution-relevant inputs for downstream cost-aware portfolio evaluation. Concretely, the layer uses a separate multi-output CatBoost regressor to forecast forward liquidity and volatility quantities at each weekly rebalance date. These forecasts are then passed into the cost-aware optimizer as point-in-time inputs for trading capacity and market impact, especially through ADV- and volatility-related measures that scale the impact term. Forecasting them in a walk-forward manner avoids relying on ex post realized values and makes the after-cost comparison across CNN, RF, and HMM portfolios more economically credible. The layer is therefore best understood as an auxiliary engineering component that strengthens the economic credibility of the main CNN/RF/HMM comparison, rather than as part of the paper’s primary predictive contribution.

F.2 Cost-aware objective, inputs, and constraints

The downstream transaction-cost layer operates in weight space rather than share space. Let w denote the post-trade portfolio weights, w^{prev} the pre-trade weights, and

$$\Delta = w - w^{prev}$$

the trade vector in weight space. Let α denote the expected-return signal, Σ the return covariance matrix, s_i the one-way linear trading-cost input for stock i , ADV_i the stock’s average daily dollar volume, σ_i a volatility input, and AUM assumed assets under management. The alpha-driven cost-aware optimizer solves a portfolio problem that trades off expected return, risk, linear trading cost, and nonlinear market impact. In economic maximization form, the problem is

$$\max_w \alpha^\top w - \lambda_{\text{risk}} w^\top \Sigma w - \eta_{\text{linear}} \sum_i s_i |\Delta_i| - \eta_{\text{impact}} \sum_i c_i |\Delta_i|^p,$$

where $p \geq 1$ is the nonlinear impact exponent and c_i is a stock-specific impact coefficient.

For numerical optimization, this is written in smooth minimization form as

$$J(w) = \lambda_{\text{risk}} w^\top \Sigma w - \alpha^\top w + \eta_{\text{linear}} \sum_i s_i \sqrt{\Delta_i^2 + \varepsilon^2} + \eta_{\text{impact}} \sum_i c_i (\Delta_i^2 + \varepsilon^2)^{p/2}.$$

The smoothing term $\varepsilon > 0$ is used only to make the absolute-value penalties differentiable and numerically stable; it does not change the underlying economic interpretation of the objective.

Under the maintained implementation assumptions, the smooth optimization problem is convex: the quadratic risk block is convex when Σ is positive semidefinite, both smoothed trading-cost terms are convex for nonnegative coefficients and $p \geq 1$, and the box, sum, and ADV-based feasibility constraints define a convex set. It follows that any local optimum is also a global optimum. In the implemented long-only alpha-driven case, the repaired covariance block is treated as positive definite in practice, so the problem is treated as strongly convex for numerical purposes and therefore has a unique global optimum. The target-tracking variant is likewise strongly convex because its quadratic block is τI with $\tau > 0$, again implying a unique global optimum.

The four terms of $J(w)$ correspond respectively to variance penalization, reward from the signal, linear spread-crossing cost, and nonlinear market-impact cost.

The stock-specific linear-cost input s_i is the one-way spread measure supplied to the optimizer. Upstream, the daily quoted one-way spread is defined as

$$s_{i,t}^{\text{raw}} = \frac{1}{2} \cdot \frac{\text{Ask}_{i,t} - \text{Bid}_{i,t}}{(\text{Ask}_{i,t} + \text{Bid}_{i,t})/2},$$

and the weekly transaction-cost pipeline constructs the optimizer input as a lagged 20-day rolling median of this raw series:

$$s_i = \text{LagMed}_{20}(s_{i,t}^{\text{raw}}).$$

Thus s_i is a smoothed point-in-time spread estimate rather than the contemporaneous daily quote. The corresponding linear trading-cost contribution for stock i is

$$\eta_{\text{linear}} s_i |\Delta_i|.$$

Let d_{exec} denote the number of trading days over which a scheduled rebalance is executed, and define the available execution-window volume as

$$V_i^{\text{exec}} = d_{\text{exec}} \text{ADV}_i.$$

The main specification sets $d_{\text{exec}} = 1$, corresponding to weekly portfolio decisions executed over one trading day. The nonlinear impact coefficient is defined as

$$c_i = \kappa_{\text{impact}} \sigma_i \sqrt{\frac{AUM}{V_i^{\text{exec}}}},$$

so that impact becomes more severe when volatility is high, liquidity is low, fund size is large, or the trade is large. The corresponding nonlinear impact contribution is

$$\eta_{\text{impact}} c_i |\Delta_i|^p.$$

The optimizer, therefore, penalizes trading more heavily in lower-liquidity, higher-volatility stocks rather than imposing a uniform cost penalty across names. Using volatility and ADV as central cost inputs is consistent with the execution literature (R. Almgren and Chriss 2001; R. Almgren, Thum, et al. 2005; Zhang et al. 2024).

Feasibility is enforced through liquidity-based bounds in weight space. The position ADV cap limits the final absolute weight using

$$\text{pos_cap}_i = \text{pos_adv_cap} \cdot \frac{\text{ADV}_i}{AUM},$$

while the trade ADV cap limits one-period trading using

$$\text{trade_cap}_i = \text{trade_adv_cap} \cdot \frac{V_i^{\text{exec}}}{AUM}.$$

The optimizer then enforces a final position bound together with a one-period trade bound. In absolute-value form, these are

$$|w_i| \leq \text{pos_cap}_i, \quad |w_i - w_i^{\text{prev}}| \leq \text{trade_cap}_i.$$

Equivalently, the trade bound can be written as

$$w_i \geq w_i^{\text{prev}} - \text{trade_cap}_i, \quad w_i \leq w_i^{\text{prev}} + \text{trade_cap}_i.$$

The position and trade bounds distinguish between final position size and one-period trade size: the former is limited by the position cap, while the latter is limited by the trade cap. Here `pos_adv_cap` and `trade_adv_cap` are user-specified liquidity-constraint parameters rather than estimated quantities. The solver also supports additional user-specified feasibility controls, including standard box constraints, long-only or dollar-neutral configurations, gross-exposure rescaling for long-short portfolios, and a partial-fill option for long-only sleeves when the requested target sum is infeasible under the ADV bounds.

F.3 Target tracking, realized costs, and numerical solution

The optimizer supports two related optimization problems. The first is the alpha-driven solver above, which chooses weights endogenously from signal strength. The second is a target-tracking solver used for strategies such as equal-weight or value-weight rebalancing. In that case, the optimizer replaces the free alpha reward with a quadratic penalty around a desired target portfolio w^{target} :

$$\min_w \frac{\tau}{2} \|w - w^{\text{target}}\|_2^2 + \eta_{\text{linear}} \sum_i s_i \sqrt{\Delta_i^2 + \varepsilon^2} + \eta_{\text{impact}} \sum_i c_i (\Delta_i^2 + \varepsilon^2)^{p/2}.$$

In implementation, this is written in the same quadratic form as the alpha-driven problem by setting

$$G = \tau I, \quad \alpha^{target} = \tau w^{target}.$$

Economically, the problem asks for the cheapest feasible way to move from w^{prev} toward the desired target portfolio, given spread, impact, and ADV constraints.

After optimization, realized trading costs are decomposed as

$$\begin{aligned} \text{linear_cost} &= \sum_i s_i |\Delta_i|, \\ \text{impact_cost} &= \sum_i c_i |\Delta_i|^p, \\ \text{total_cost} &= \text{linear_cost} + \text{impact_cost}. \end{aligned}$$

Separating the two components helps distinguish ordinary spread-crossing costs from the additional market-impact cost associated with trading larger sizes in less liquid names.

From a numerical perspective, the alpha-driven solver uses projected L-BFGS with Armijo backtracking, whereas the target-tracking solver uses projected gradient descent with a Barzilai–Borwein step-size initialization and Armijo safeguarding. These differences reflect numerical optimization choices rather than changes in the underlying economic model: the objective, trading-friction terms, and feasibility projection remain unchanged, while the more strongly curved alpha-driven problem is handled by a solver better suited to that geometry.

F.4 Data and feature construction

Feature construction uses the cleaned daily auxiliary parquet, with the weekly backbone serving primarily to define the weekly rebalance rows to which daily quantities are aligned. The daily feature builder standardizes the panel on $(Date, StockID)$, validates the core inputs `Close`, `Vol`, and `Ret`, and constructs derived variables including dollar volume, cleaned returns, market-cap proxies, turnover where shares are available, and log-transformed price, dollar volume, and market capitalization. It then adds stock-level rolling features over 5-, 20-, and 60-day windows, including means and standard deviations of log dollar volume, realized volatility, average absolute returns, momentum, and turnover, together with simple calendar indicators. Daily features are lagged centrally in the dataset builder, and optional weekly return features from the backbone are shifted by one weekly row to preserve a clear point-in-time convention.

F.5 Auxiliary liquidity and volatility targets

The transaction-cost forecasting layer is trained on forward-looking measures of average dollar volume and return volatility constructed from the cleaned daily auxiliary data. These targets are stored both in logs and in levels as `target_log_adv`, `target_log_vol`, `target_adv`, and `target_vol`. In the default specification, they are formed over a five-trading-day forward window beginning on the row date. The weekly backbone provides only the rebalance keys used to align the daily targets with the weekly portfolio pipeline.

F.6 Forecasting model and walk-forward protocol

The forecasting model is a thin wrapper around a multi-output `CatBoostRegressor` trained on the log targets. In the default specification, the model uses the `MultiRMSE` loss with 600 boosting iterations, a learning rate of 0.05, a depth of 6, an L_2 regularization of 3.0, Bernoulli subsampling of 0.8, feature subsampling of 0.8, a minimum leaf size of 100, and a random seed of 7. The wrapper stores the fitted feature and target schema and saves both the trained model and associated metadata.

Out-of-sample generation follows an annual expanding-window design. The runner first builds the weekly-aligned modeling panel and then, for each out-of-sample year, fits the CatBoost model on all prior years and evaluates it on the target year. Predictions are generated in log space and transformed back to levels where needed. The procedure also records annual diagnostics, including RMSE, MAE, and correlation for the log targets. The overall workflow is annual, walk-forward, and leakage-aware.

F.7 Forecasting diagnostics

Table 3: Period means of annual out-of-sample CatBoost forecasting diagnostics

Period	Mean n	Corr. ADV	Corr. vol	RMSE ADV	RMSE vol
2000–2024	333,848	0.976	0.760	0.568	0.642
2016–2024	391,849	0.973	0.832	0.624	0.597
2016–2019	322,840	0.978	0.813	0.555	0.583
2020–2024	447,057	0.970	0.846	0.680	0.609

Table 3 reports out-of-sample forecasting diagnostics for the transaction-cost layer, averaged over the reported subperiods. For each target, the reported correlation is the correlation between the predicted and realized log target, and the reported RMSE is the root mean squared prediction error in that same realized log target. The strongest result is the forecast of future log average dollar volume, for which the CatBoost model delivers consistently high predicted-versus-realized correlations across the sample. Forecasts of future log volatility are weaker than the ADV forecasts over the full period, but are materially stronger in 2016–2024 and strengthen further in 2020–2024. The later subperiod is also associated with a modest increase in log-ADV RMSE, even though the ADV correlation remains very high and the evaluation set becomes substantially larger. Overall, the diagnostics suggest that the model provides a strong source of forward-looking liquidity inputs and a reasonably informative source of forward-looking volatility inputs for downstream cost-aware portfolio evaluation.

F.8 Interpretation

The transaction-cost forecasting layer is included to improve implementation realism rather than to alter the paper’s central empirical question. Its role is to provide forward-looking liquidity and volatility inputs for downstream portfolio evaluation in a leakage-aware manner, thereby improving the economic credibility of after-cost portfolio comparisons without changing the substantive comparison: whether structured context improves a chart-based CNN and whether state-sensitive fusion improves on either standalone branch. The broader rationale is well established in the execution literature (Perold 1988; R. Almgren and Chriss 2001; Patton and Weller 2020; Goyenko et al. 2024). In the present setting, these forecasts are incorporated into a weekly return-prediction pipeline, enabling more realistic evaluation of after-cost portfolio performance. For that reason, the material is reported in the supplementary appendix rather than the main text. Related empirical interpretation of the screened-versus-unscreened comparison is reported in Appendix G.10.

G Diagnostic interpretation and robustness

G.1 Gross return spanning and signal persistence

The gross-performance comparisons are supplemented with formal inference and diagnostics of signal stability. Pairwise Sharpe differences are evaluated using a paired circular block bootstrap. Each replication resamples consecutive eight-week blocks with replacement, using the same resampled weeks for both strategies in a comparison. We use 5,000 replications and report 95% percentile confidence intervals. The procedure is conditional on the previously generated out-of-sample return streams; the underlying prediction models are not retrained within each bootstrap replication.

Table 4: Paired block-bootstrap inference for gross Sharpe differences. Differences are HMM minus the indicated benchmark.

Benchmark	ΔSR	95% CI	$\Pr(\Delta SR > 0)$
RF	+0.319	[+0.191, +0.453]	1.000
50/50 rank blend	-0.249	[-0.450, -0.049]	0.007
Logistic stack	-0.035	[-0.130, +0.057]	0.217
Expected-return fusion	-0.017	[-0.077, +0.039]	0.276

The bootstrap results show that HMM reliably improves on RF in gross risk-adjusted performance, but does not outperform the simpler 50/50 rank blend. Its Sharpe differences relative to the logistic stack and expected-return fusion are small and statistically indistinguishable from zero.

Return-spanning regressions provide a complementary test of whether HMM contains return information beyond that already captured by the benchmark strategies. For benchmark return vector R_t^B , we estimate

$$R_t^{HMM} = \alpha + \beta^\top R_t^B + \varepsilon_t.$$

The intercept α measures the average component of HMM returns not explained by the benchmark return stream. A statistically significant positive intercept therefore indicates return information not spanned by the benchmark. Regressions use weekly common-universe high-minus-low returns, with HAC/Newey–West inference; reported alphas are annualized.

Table 5: Gross return-spanning regressions for HMM. The dependent variable is the weekly HMM high-minus-low return. Alphas are annualized; confidence intervals and p -values use HAC/Newey–West inference.

Benchmark	Ann. α	95% CI	p -value	R^2
RF	+11.77%	[+6.82%, +16.73%]	< 0.001	0.913
CNN	+105.35%	[+95.09%, +115.62%]	< 0.001	0.009
RF + CNN	+3.74%	[+1.25%, +6.22%]	0.003	0.955
50/50 rank blend	+1.34%	[-3.40%, +6.08%]	0.580	0.880
Logistic stack	+0.87%	[-1.55%, +3.29%]	0.480	0.969
Expected-return fusion	+0.28%	[-1.04%, +1.61%]	0.677	0.988
RF + logistic stack	+0.47%	[-1.06%, +2.00%]	0.548	0.988

The spanning results separate the value of fusion from the value of the HMM specifically. Relative to RF alone, HMM produces an economically and statistically significant annualized alpha of 11.77%. A significant 3.74% alpha remains when both RF and CNN returns are included as benchmarks, indicating that the HMM return stream is not simply a linear combination of the two standalone strategies. In contrast, HMM alpha is small and statistically insignificant relative to the rank blend, logistic stack, and expected-return fusion. The evidence therefore supports the value of combining complementary signals, but does not establish latent-state switching as a distinct source of gross alpha.

Signal-persistence diagnostics examine how much each model’s cross-sectional ranking and extreme-tail membership change from one week to the next. Rank persistence is the week-to-week Spearman correlation of stock ranks among names observed in both weeks. Tail retention is the fraction of stocks in the previous week’s long or short decile that remain in the same decile one week later. Mean holding length measures the average number of consecutive weeks that a selected stock remains in the corresponding tail.

Table 6: Signal-persistence diagnostics. Tail retention is the fraction of the previous week’s selected names retained in the same sleeve. Holding length is the mean consecutive number of weeks that a stock remains selected.

Model	Rank persistence	Long retention	Short retention	Long hold	Short hold
CNN	0.022	10.6%	10.5%	1.12	1.12
RF	0.810	65.5%	66.2%	2.93	2.99
50/50 rank blend	0.365	22.6%	25.0%	1.29	1.34
Logistic stack	0.527	36.7%	37.9%	1.58	1.62
HMM	0.651	46.9%	47.9%	1.90	1.94
Expected-return fusion	0.602	42.6%	43.6%	1.75	1.78

The persistence diagnostics place HMM between RF and the simpler fusion rules. RF produces the most stable rankings and retains roughly two-thirds of both tail portfolios from one week to the next. HMM is the most persistent fusion method, with rank persistence of 0.651, followed by expected-return fusion at 0.602, logistic stacking at 0.527, and the 50/50 rank blend at 0.365. HMM also retains approximately 47% of each tail and has mean holding spells of about 1.9 weeks, compared with roughly 1.8 weeks for expected-return fusion, 1.6 weeks for logistic stacking, and 1.3 weeks for the rank blend. HMM therefore gives up some of RF’s persistence while filtering more week-to-week variation than the alternative fusion rules. These differences in signal stability provide a direct link between the gross signal results and the implementation and capacity analysis that follows.

G.2 Fixed-AUM net performance and capacity

Table 7 reports net Sharpe across the fixed-AUM grid. HMM leads at \$10 million and \$25 million, while RF leads from \$50 million onward. The rank blend illustrates the gap between gross and implementable performance: despite having the highest gross Sharpe, its net Sharpe deteriorates much more rapidly with AUM.

Table 7: Fixed-AUM net Sharpe.

AUM	RF	Rank blend	Logistic	HMM	Exp.-return fusion
\$10m	3.464	3.731	3.690	3.850	3.446
\$25m	3.265	3.256	3.290	3.414	3.229
\$50m	3.030	2.796	2.888	3.000	2.512
\$100m	2.939	2.098	2.396	2.628	2.357
\$250m	2.410	0.999	1.748	2.088	1.843
\$500m	2.002	0.464	1.302	1.778	1.400
\$1bn	1.726	0.247	1.174	1.396	1.204

Table 8: Paired block-bootstrap inference for net Sharpe differences at \$100 million AUM. Differences are HMM minus the indicated benchmark.

Benchmark	ΔSR	95% CI	$\Pr(\Delta SR > 0)$
RF	-0.311	[-0.508, -0.121]	0.001
50/50 rank blend	+0.530	[+0.280, +0.797]	1.000
Logistic stack	+0.232	[+0.044, +0.423]	0.992

At \$100 million, paired bootstrap inference gives HMM-minus-RF $\Delta SR = -0.311$ with a 95% interval of approximately $[-0.508, -0.121]$. HMM remains significantly above rank blend and logistic stacking at that AUM. The fixed-AUM grid should be interpreted as a set of tested scale points rather than as a continuous capacity estimate.

At \$1 billion, executed gross Sharpe remains higher for HMM (2.856) than for RF (2.701), even though net Sharpe reverses to 1.396 for HMM versus 1.726 for RF.

Hard capacity constraints become important only after the initial RF–HMM crossover. Around \$50–\$100 million, position- and trade-cap binding is still limited, so nonlinear market impact creates a soft-capacity region before constraints materially reshape the portfolio. At \$1 billion, position caps bind on nearly all dates for both models, while trade caps bind more frequently for HMM (about 73.7% of dates) than RF (about 62.5%). This is consistent with HMM wanting to reposition more frequently rather than simply holding systematically larger positions.

G.3 Turnover, nonlinear impact, and the short-side bottleneck

The direct RF–HMM cost decomposition indicates that HMM’s higher implementation burden is almost entirely a turnover-quantity effect. At approximately \$100 million, RF’s average cost is about 47.5 bps per rebalance and HMM’s about 76.4 bps. A symmetric quantity–intensity decomposition attributes essentially the entire difference to greater turnover; the average cost-intensity contribution is slightly negative. Figure 4 shows that this turnover-quantity effect dominates across the full AUM grid. Direct cost components show that roughly 90% of the incremental HMM burden comes from nonlinear impact rather than the linear spread term. Liquidity characteristics of RF and HMM portfolios are otherwise similar, so the evidence does not support a simple story in which HMM systematically selects much less liquid stocks.

Table 9: RF–HMM trading-cost diagnostics at \$100 million. Panel A decomposes the difference in average weekly trading costs. Panel B compares execution characteristics across HMM’s long and short sleeves.

<i>Panel A: RF–HMM cost decomposition</i>				
Metric	RF	HMM	HMM–RF	Units
Average trading cost per rebalance	47.5	76.4	+28.9	bps
Total turnover	0.570	0.979	+0.409	–
Turnover-quantity contribution	–	–	+29.11	bps
Cost-intensity contribution	–	–	-0.17	bps
Linear spread contribution	–	–	+2.86	bps
Nonlinear impact contribution	–	–	+26.08	bps

<i>Panel B: HMM sleeve-level execution characteristics</i>				
Metric	Long sleeve	Short sleeve	Short/long	Units
Trading cost relative to turnover	50.07	102.00	2.04×	bps
Average spread	4.36	5.70	1.31×	bps
Avg. forecast stock-level ADV	202.1	130.5	0.65×	\$m/day
Avg. forecast daily volatility (5-day window)	1.18%	2.02%	1.71×	–
Share of incremental HMM–RF cost	32.5%	67.5%	2.08×	–

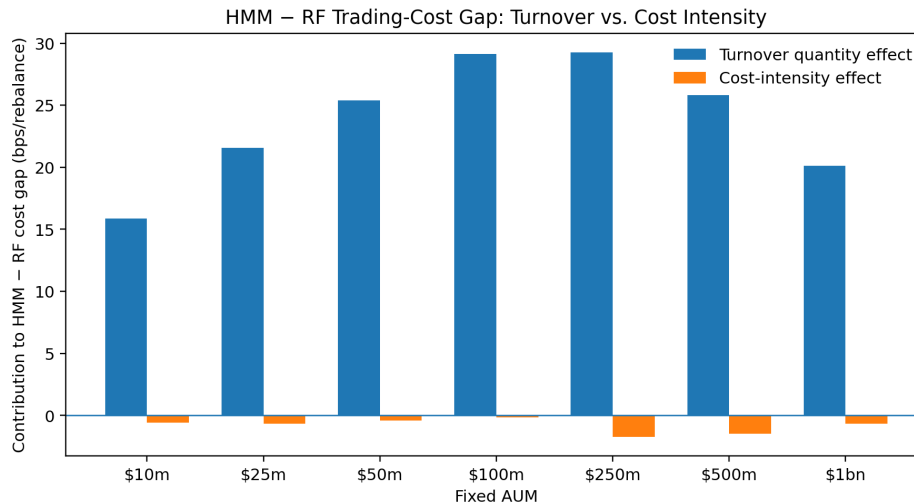


Figure 4: Decomposition of the HMM-minus-RF trading-cost gap across fixed AUM. The higher cost of HMM is attributable almost entirely to greater turnover rather than higher average cost intensity.

Sleeve-level diagnostics reveal a strong asymmetry. At \$100 million, HMM’s short-side trading cost per unit of turnover is roughly twice its long-side value. The short sleeve also has lower ADV, wider spreads, and higher volatility, and accounts for roughly two-thirds of HMM’s incremental cost relative to RF. The same short-side asymmetry is present in RF.

G.4 Risk decomposition and sector structure

HMM is less sector concentrated than RF throughout the tested AUM grid. At \$100 million, mean sleeve sector HHI is 0.219 for HMM and 0.261 for RF. The variance decomposition provides a separate accounting of the portfolio-risk difference. Table 10 shows that the short sleeve is the dominant source of RF’s higher after-cost portfolio variance at every tested AUM, while the long-sleeve and covariance contributions are considerably smaller and can change sign. Sector diversification is therefore consistent with HMM’s lower risk, but the exact decomposition identifies the short sleeve as the direct source of most of the variance difference.

Table 10: Sector concentration and RF–HMM after-cost risk decomposition across fixed AUM. Panel A reports mean sleeve sector HHI and annualized net long–short volatility. Panel B decomposes the RF-minus-HMM difference in annualized return variance. Variance contributions are multiplied by 10^4 for readability; positive values increase RF variance relative to HMM.

<i>Panel A: Sector concentration and net volatility</i>				
AUM	HMM HHI	RF HHI	HMM vol.	RF vol.
\$10m	0.223	0.268	24.04%	26.99%
\$25m	0.222	0.269	24.13%	26.55%
\$50m	0.222	0.266	24.51%	26.89%
\$100m	0.219	0.261	24.64%	26.33%
\$250m	0.215	0.252	24.52%	27.03%
\$500m	0.202	0.243	24.86%	27.44%
\$1bn	0.184	0.216	24.74%	26.66%

<i>Panel B: Contribution to RF-minus-HMM variance difference</i>				
AUM	Long sleeve	Short sleeve	– 2 covariance	Total
\$10m	-9.44	+151.90	+8.23	+150.69
\$25m	-23.82	+139.65	+6.87	+122.70
\$50m	-5.66	+138.29	-10.18	+122.44
\$100m	+4.15	+92.44	-10.38	+86.21
\$250m	-7.92	+118.98	+18.36	+129.41
\$500m	+3.46	+147.57	-16.05	+134.97
\$1bn	+12.56	+113.97	-27.81	+98.72

Table 11: Exact RF–HMM variance decomposition at \$100 million. Portfolio variance is decomposed as $\text{Var}(L - S) = \text{Var}(L) + \text{Var}(S) - 2\text{Cov}(L, S)$, where L and S denote long- and short-sleeve returns. The final column reports the RF minus HMM difference for each component.

Component	RF	HMM	RF – HMM
Long-sleeve variance	296.78	292.63	+4.15
Short-sleeve variance	895.62	803.18	+92.44
– 2 $\times$ covariance term	-499.07	-488.69	-10.38
Total portfolio variance	693.34	607.13	+86.21

Note: Annualized variance contributions are multiplied by 10^4 for readability. The decomposition shows that the RF–HMM variance difference is driven overwhelmingly by the short sleeve.

G.5 Latent-state remapping rather than return timing

The mechanism tests distinguish return timing from portfolio remapping. Table 12 shows no statistically significant relationship between HMM-minus-logistic returns and state entropy, maximum state probability, or transitions in the dominant state. State changes are instead strongly associated with portfolio movement. A dominant-state transition is associated with approximately +0.38 target turnover, where target turnover measures changes in the desired long and short portfolio weights before transaction costs and trading constraints are applied. A post-hoc attribution proxy is constructed each week by regressing HMM forecast logits cross-sectionally on standardized CNN and RF logits and normalizing the mean absolute fitted contributions to sum to one. Absolute week-to-week changes in the resulting CNN attribution share explain about 22% of HMM target-turnover variation.

The relationship remains after implementation. At \$100 million, a dominant-state transition is associated with approximately +0.29 realized turnover and a 12.5 percentage-point increase in the average fraction of names replaced across the long and short sleeves. The diagnostics support interpreting the latent states as changes in the fusion mapping rather than as successful timing of incremental HMM returns.

Table 12: HMM state-mechanism diagnostics. Return regressions use weekly HMM-minus-logistic returns. Portfolio-movement regressions use HMM target portfolios or the implemented \$100 million portfolio. Inference uses HAC/Newey–West standard errors.

Outcome	Explanatory variable	Coefficient	HAC t	p -value
<i>Panel A: Incremental return timing</i>				
HMM–logistic return	State entropy	-0.1464%	-0.601	0.548
HMM–logistic return	Max. state probability	-0.0421%	-0.186	0.852
HMM–logistic return	State transition	-0.0932%	-0.912	0.362
<i>Panel B: Portfolio remapping</i>				
Target turnover	State transition	+0.378	12.033	< 0.001
Target turnover	Abs. change in CNN attribution share	+4.130	14.944	< 0.001
Realized turnover, \$100m	State transition	+0.286	5.985	< 0.001
Avg. sleeve replacement, \$100m	State transition	+0.125	4.977	< 0.001

Target turnover is measured before transaction costs and trading constraints and sums turnover across the long and short sleeves; a change of 1 corresponds to turnover equal to the full notional of one sleeve. Realized turnover is measured from the implemented portfolio after the cost-aware trading step. Average sleeve replacement is the mean of the long- and short-sleeve replacement fractions. The regression of target turnover on the absolute weekly change in the post-hoc CNN attribution share has $R^2 = 0.222$. Separately, the sleeve-replacement coefficient of 0.125 corresponds to a 12.5 percentage-point increase.

G.6 Diagnostic sleeve swaps

To isolate the contribution of each side of the portfolio, we construct two diagnostic sleeve swaps: one combines HMM’s long sleeve with RF’s short sleeve, while the other combines RF’s long sleeve with HMM’s short sleeve. The sleeves are taken directly from the saved implemented portfolios and are not jointly reoptimized.

Table 13 shows that the short sleeve carries much of the RF–HMM after-cost performance difference at moderate and high AUM. At \$100 million, combining HMM’s long sleeve with RF’s short sleeve produces a Sharpe ratio close to RF, while the opposite combination produces a Sharpe ratio close to HMM. The matching is nearly exact at \$1 billion. At \$25 million, however, the HMM long sleeve still contributes additional performance, so the low-AUM HMM advantage is not exclusively a short-side effect.

Table 13: Diagnostic sleeve-swap results. Each hybrid is compared with the original strategy supplying its short sleeve. Confidence intervals are from the paired eight-week block bootstrap with 5,000 replications.

AUM	Portfolio	Sharpe	Benchmark	ΔSR	95% CI
\$25m	HMM long + RF short	3.429	RF: 3.265	+0.164	[+0.062, +0.272]
\$25m	RF long + HMM short	3.271	HMM: 3.414	-0.143	[-0.260, -0.030]
\$100m	HMM long + RF short	2.967	RF: 2.939	+0.028	[-0.059, +0.114]
\$100m	RF long + HMM short	2.609	HMM: 2.628	-0.019	[-0.107, +0.070]
\$1bn	HMM long + RF short	1.725	RF: 1.726	-0.001	[-0.065, +0.060]
\$1bn	RF long + HMM short	1.403	HMM: 1.396	+0.007	[-0.060, +0.075]

G.7 Dynamic NAV, missing returns, and numerical robustness

Table 14 summarizes three implementation robustness exercises. First, dynamic-NAV simulations starting from \$100 million produce substantially lower Sharpe ratios than the corresponding fixed-AUM tests as portfolio growth feeds back into market impact and capacity constraints. These paths are interpreted as compounding stress tests rather than forecasts of terminal wealth.

Second, the qualitative RF–HMM ranking is robust to severe missing-return assumptions. Even when the first missing return in each spell is assigned a 100% loss, HMM remains ahead at \$25 million and RF remains ahead at \$100 million and \$1 billion, although the \$100 million difference becomes very small. The exercise therefore supports the broad AUM-dependent ranking while showing that performance levels near the crossover are sensitive to the missing-return convention.

Finally, a targeted convergence audit re-solved sampled problematic optimization calls at 2,000, 4,000, and 10,000 iterations. Objectives and portfolio weights were numerically stable across the larger iteration budgets,

supporting the interpretation of the frequent production convergence flags as a stopping-diagnostic issue rather than economically unfinished portfolio solutions.

Table 14: Implementation and numerical robustness diagnostics. Panel A compares the \$100 million fixed-AUM benchmark with a dynamic-NAV simulation initialized at \$100 million.

<i>Panel A: Fixed AUM versus dynamic NAV</i>				
Model	Fixed-AUM SR	Dynamic-NAV SR	Max drawdown	Avg. turnover
RF	2.939	1.329	-26.35%	0.214
HMM	2.628	1.270	-28.09%	0.314
<i>Panel B: Severe missing-return stress</i>				
AUM	RF baseline	RF stressed	HMM baseline	HMM stressed
\$25m	3.265	3.199	3.414	3.374
\$100m	2.939	2.578	2.628	2.567
\$1bn	1.726	1.623	1.396	1.369
<i>Panel C: Optimizer convergence audit</i>				
Audit	Models	AUM	Iteration budgets	Result
Exact-problem re-solves	RF, HMM	\$100m	2k / 4k / 10k	Stable

G.8 HMM tuning and fit diagnostics

The selected HMM remains numerically well behaved in its estimation diagnostics. Candidate rankings, validation-year results, EM likelihood progression, transition probabilities, and filtered state probabilities are useful for establishing that the fitted latent-state system is persistent but non-degenerate. These diagnostics should be interpreted as evidence about numerical and statistical behavior, not as proof that the states correspond to structural economic regimes.

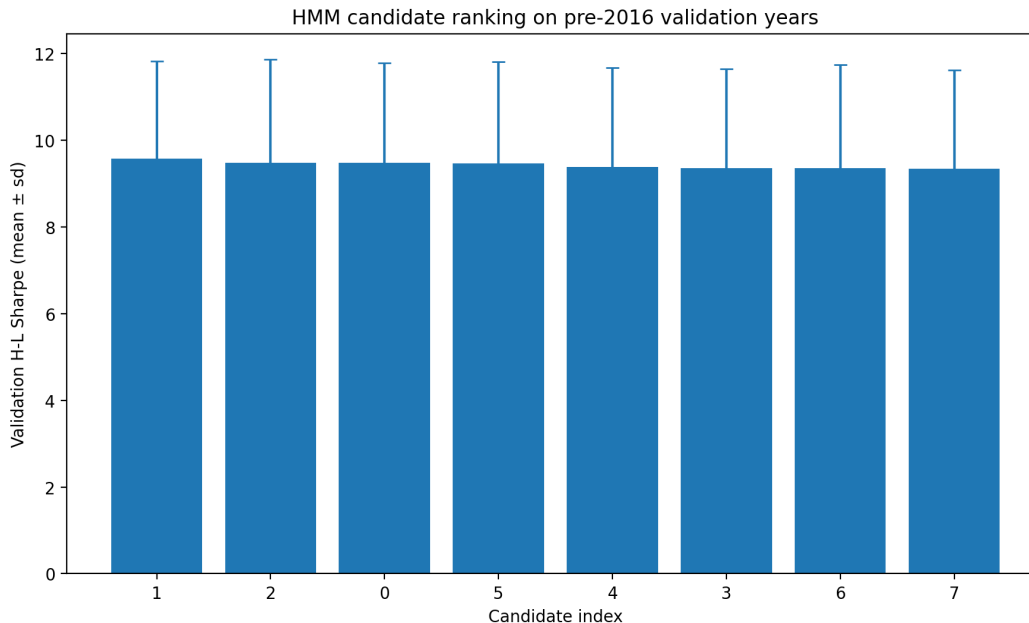


Figure 5: Ranking of HMM candidate specifications on the pre-2016 validation exercise.

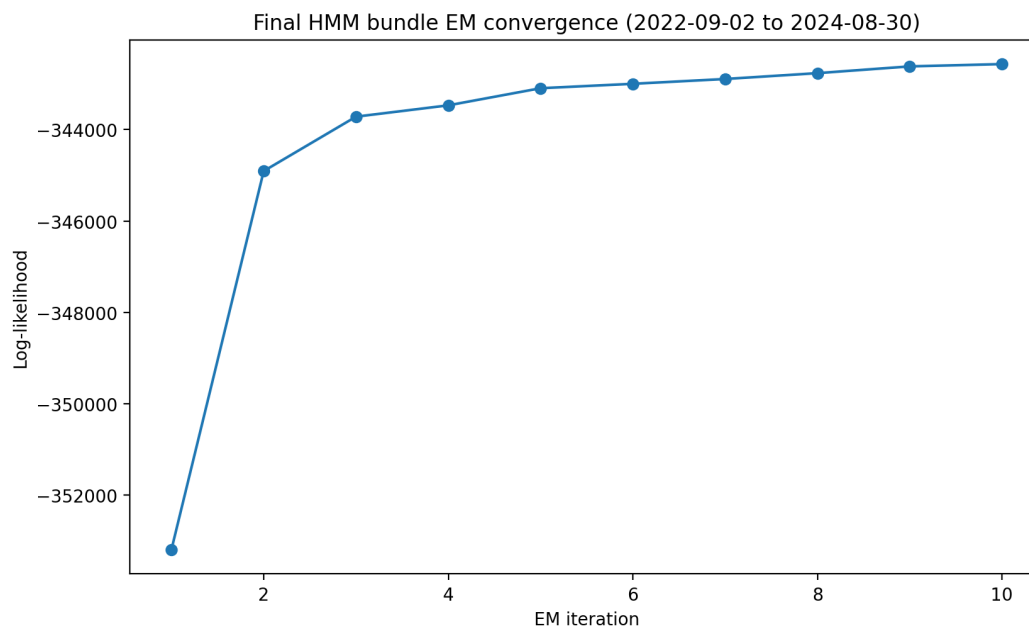


Figure 6: EM convergence for the final stored HMM bundle.

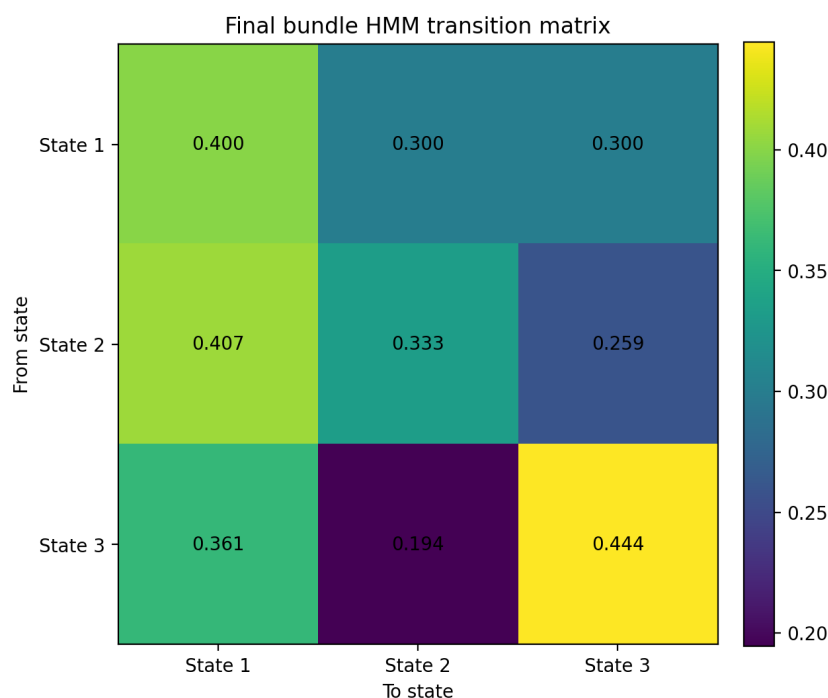


Figure 7: Estimated transition matrix for the selected HMM. The matrix is persistent but not degenerate.

G.9 Final-bundle state interpretation

The final fitted HMM bundle uses three latent states with state-specific logistic emission blocks. Because this bundle represents only the terminal refit rather than the full sequence of historical walk-forward fits, individual state labels should be treated as descriptive rather than structural. Coefficient patterns differ across states, but the

full-sample remapping diagnostics in Appendix G.5 provide stronger evidence about the economic role of the latent states. We therefore avoid assigning literal economic regime labels to individual states.

G.10 Screened versus unscreened robustness

The earlier screened-versus-unscreened exercise remains useful as a secondary tradability robustness check. Unscreened gross performance is stronger in several specifications but erodes sharply after estimated trading frictions, while screening shifts the portfolios toward more liquid and implementable names. The result supports the use of tradability controls, but it is no longer the central economic comparison because the fixed-AUM stateful implementation provides a more direct test of cost and capacity.

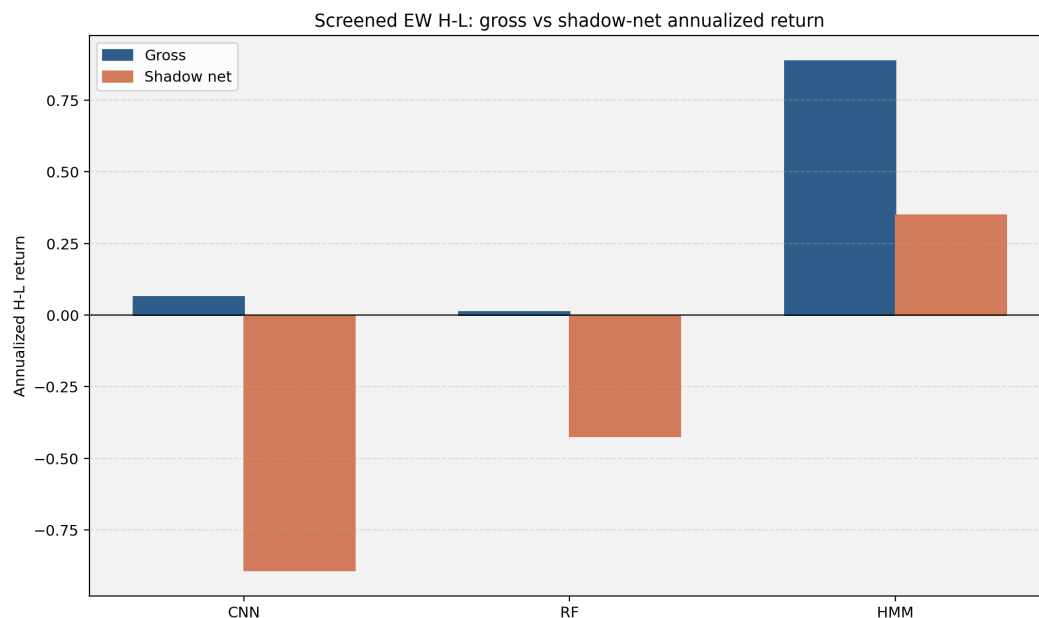


Figure 8: Legacy screened equal-weight H–L performance before and after the shadow transaction-cost adjustment. This robustness exercise is supplementary to the stateful fixed-AUM implementation.